

COMP 3602 : Mini Project I

Fall 2026

Sampling Techniques + Simple Visualization

Scenario

A retail company wants to understand the satisfaction of its customers across six branches. For this lab, the complete CSV will be treated as the known customer population. You will draw different samples, compare their estimates with the population, and use simple plots to investigate the data.

Central question

What is the average customer satisfaction score across the company's customers?

Dataset

Variable	Meaning
Customer_ID	Unique customer identifier
Branch	Retail branch visited
Region	Geographical region
Customer_Type	Regular, Premium, or Business
Age	Customer age
Waiting_Time_Min	Waiting time in minutes
Purchase_Amount_OMR	Purchase amount in OMR
Satisfaction_Score	Customer satisfaction score from 1 to 5

Mission 1 - LOAD and INSPECT the population

1. Load the CSV file into a DataFrame called customers.

```
import pandas as pd

customers = pd.read_csv("Retail_Customers.csv")
```

2. Display the first five rows.

3. Find the number of rows and columns.

4. Display the column names and inspect the data types.

5. How many customers belong to each branch?

GO FURTHER

Which customer type is most common? What percentage of the population does it represent?

Mission 2 - Know the population before sampling

6. Calculate the population mean of Satisfaction_Score.
7. Calculate the population mean of Waiting_Time_Min.
8. Calculate the population mean of Purchase_Amount_OMR.

Think before moving on

In a real study, would we normally know these population values? Why are we able to know them in this synthetic exercise?

Mission 3 - Simple Random Sampling

9. Randomly select 100 customers from the population.

```
random_sample = customers.sample(n=100, random_state=42)
```

10. Calculate the sample mean satisfaction. How close is it to the population mean?
11. Create another random sample of 100 customers using a different random_state. Is the answer exactly the same?
12. Explain in one sentence why two random samples can give different estimates.

GO FURTHER

Repeat with sample sizes 50, 100 and 300. Which sample size tends to give a more stable estimate of the population mean?

Mission 4 - Systematic Sampling

13. Starting at row 5, select every 15th customer.

```
start = 5  
systematic_sample = customers.iloc[start::15]
```

14. Find the mean satisfaction of the systematic sample.
15. Compare it with the population mean and the simple random sample mean.

16. In words, describe how the systematic sample was selected.

Mission 5 - Stratified Sampling

17. Treat each Branch as a stratum.

18. Create one DataFrame for each branch using filtering.

19. Randomly select 15 customers from EACH branch.

20. Combine the six small samples using `pd.concat()`.

21. Confirm that every branch is represented in the final sample.

22. Calculate its mean satisfaction and compare it with the population mean.

Remember

STRATIFIED = take SOME customers from EVERY branch.

Mission 6 - Cluster Sampling

23. Select two whole branches, for example Sohar and Nizwa.

```
selected_branches = ["Sohar", "Nizwa"]

cluster_sample = customers[
    customers["Branch"].isin(selected_branches)
]
```

24. Calculate the mean satisfaction of this cluster sample.

25. Compare cluster sampling with stratified sampling. What was selected in each case?

Remember

CLUSTER = select SOME whole groups. STRATIFIED = sample SOME individuals from EVERY group.

Mission 7 - Create a biased sample

26. Imagine a researcher surveys only customers who waited less than 5 minutes.

27. Create this sample and calculate its mean satisfaction.

28. Compare it with the population mean.

29. Is Python wrong if the estimate is different? Explain where the problem occurs.

Key idea

Correct calculation does not guarantee a valid conclusion if the sample itself is biased.

Mission 8 - VISUALIZE what sampling did

30. Create a Series containing the mean satisfaction from the Population, Random, Systematic, Stratified, Cluster and Biased samples.

31. Use Matplotlib to create a bar chart comparing these values.

```
import matplotlib.pyplot as plt

plt.bar(results.index, results.values)
plt.xlabel("Sampling Method")
plt.ylabel("Average Satisfaction Score")
plt.title("Same Population, Different Samples")
plt.xticks(rotation=30)
plt.show()
```

32. Which sampling method gives the most noticeably different estimate? Explain why.

33. Does the bar closest to the population automatically prove that the sampling method is always the best? Explain briefly.

Mission 9 - Independent investigation

Your investigation

A manager says: "Customers who spend more money are more satisfied." Investigate whether the data supports this statement.

34. Identify the two variables you need.

35. Choose an appropriate plot yourself. A scatter plot is suitable for investigating two numerical variables.

36. Produce the plot with a suitable title and axis labels.

37. Write 2-3 sentences interpreting what you observe. Use the word association rather than causation.

GO FURTHER

Create one additional question of your own using this dataset. Choose an appropriate plot and provide a short evidence-based interpretation.

Submit:

1. Your notebook containing the code, outputs, plots and short interpretations.

2. Save using this format: YourID_FirstName.ipynb in the moodle page.